

10/7/23
EEE

SASTRA DEEMED UNIVERSITY
(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

JULY 2023

Course Code: **EEE101**

Course: **BASIC ELECTRICAL ENGINEERING**

QP No.: **UF006-2**

Duration: **3 hours**

Max. Marks: **100**

PART-A

10 x 2 = 20 Marks

Answer all questions

1. State Kirchhoff's current law and voltage law.
2. Why domestic appliance is connected parallel?
3. A voltage $e = 200 \sin \omega t$ when applied to a resistance is found to give a power 100 watts. Find the value of resistance and the equation for current.
4. A wire having a length of 1 m moves at right angles to its length at 50 m/sec in a uniform magnetic field of 1 wb/m^2 . Determine the emf induced the conductor when the direction of motion is (a) at right angles to the field (b) inclined at 30° to the direction of the field.
5. Explain Fleming's Right-Hand Rule.
6. Explain coupling coefficient in magnetic circuit.
7. Can DC voltage apply to a transformer?
8. State faraday's law of electromagnetic induction.

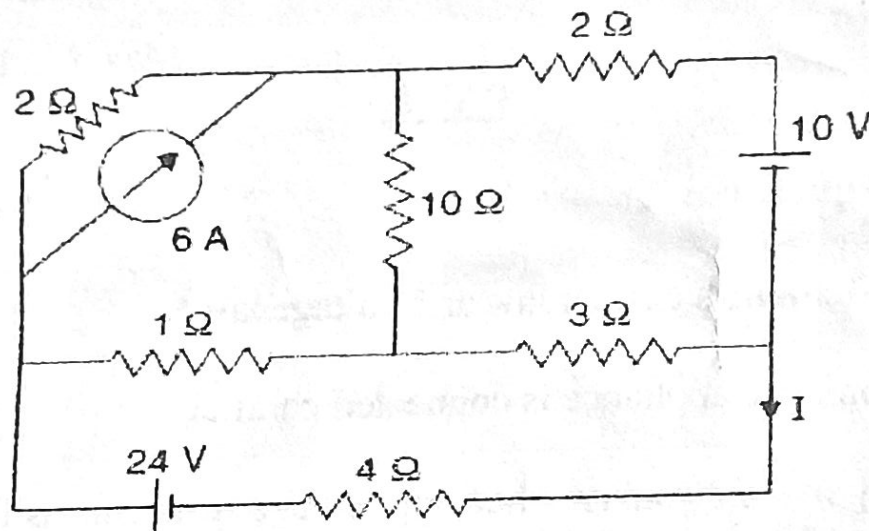
9. Explain the importance of Earthing in electrical wiring.
10. List any four advantages of three phase system.

PART-B

Answer any FOUR questions

4 x 15 = 60 Marks

11. Determine the current in the 4-ohm resistance of the circuit shown in figure.



12. a) Derive the expression for RMS value, Average value and form factor of a sinusoidal alternating current. (10)
- b) Compare electric field and magnetic field. (5)

13. Explain the construction and working principle of DC Generator.

14. a) State Faraday's Law of Electromagnetic Induction and Statically Induced emf. (5)
- b) The number of turns in a coil is 250. When a current of 2A flows in this coil. The flux in the coil is 0.3 mwb. When this current is reduced to zero in 2 milliseconds, the voltage induced in a coil lying in the vicinity of coil is 63.75 volts. Find the inductances the

$$L = 0.75$$

first, second coil and the number of turns of the second coil. (10)

15. a) Derive the emf equation of a single-phase transformer. (5)

b) What are wiring materials that are used in house wiring? (10)

16. A small house is having following appliances. Determine the monthly Electricity bill, if the electric tariff is Rs 8/- per unit

Name of the appliance	Power consumption	Daily usage in Hours
4 LED bulbs	9-W each	10
2 FANS	60-W each	16
1 TV	100-W	5
1 Refrigerator	200-W	10

PART-C

Answer the following

1 x 20 = 20 Marks

17. a) A resistive network has $AB = 9 \Omega$; $BC = 1 \Omega$; $CA = 1.5 \Omega$ forming a delta and $AD = 6 \Omega$; $BD = 4 \Omega$ and $CD = 3 \Omega$ forming a star.

Compute the network resistance measured between (i) A and B (ii) B and C (iii) C and A.

b) Explain the importance of starters in D.C motors.

c) Define and derive the equation for voltage regulation in a Transformer.

d) Draw the one-line diagram of Generation, Transmission and Distribution system.
